

Installing and running NAMD

Case: Mahti

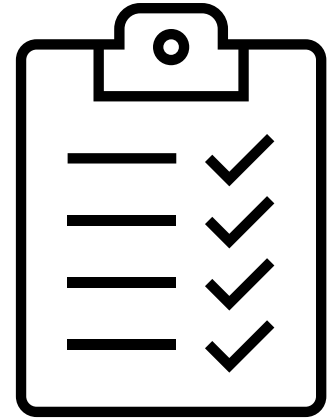


Downloading NAMD

- Normally for downloading, you need to create a user at the distribution site and agree to a user agreement
 - Today, you can just copy the NAMD .tar file in our project folder
- Many different versions available, 3.0 is the one we will be using
 - There is a Mahti module for NAMD 2.14 already
- For our installation, we will download the source code and configure our compilation ourselves
- For local testing on your Linux machine, you can download a readymade binary (e.g. Linux-x86_64-multicore)

NAMD dependencies

- Charm++
 - "Alternative" to MPI
 - Needs to be compiled against MPI before compiling NAMD
- FFTW and tcl libraries
 - We will use a ready binary for both



Compilation options in NAMD

- SMP vs no-SMP for Charm++
 - Shared memory parallelism
 - User-specified number of CPUs reserved for communication only
 - SMP might help in scaling the program (>2 nodes)
- Compiler flags
 - Choosing the right ones can improve the performances you get significantly

Running NAMD

- We are running NAMD on Mahti
 - Example Slurm scripts can be based off of examples from the CSC Docs
- We will do a minimal scaling exercise today (16 cores vs 32 cores)
- NAMD input file contains tuning options you might be able to change (sometimes at the expense of accuracy)
 - Cutoff and switchdist
 - e.g. cutoff 12 Å -> 9 Å
 - And switchdist 10 Å -> 8 Å
 - Choosing a larger timestep
 - Choosing larger "outputtiming" and "outputEnergies" values to minimize I/O
 - Check the NAMD documentation for details

Analyzing the results

- NAMD puts out two values that are of interest to us:
 - WallClock (Total time taken to run the simulation)
 - Benchmark time (days/ns)
- Days/ns is more easily interpreted when you switch it to "ns per day"
 - Ready python script for this in the exercise directory
- Use the "ns per day" value to gauge the performance
 - Higher values equal better performances

Visualizing our simulation

- Lastly, you should visualize the Apoa1 protein to know what you're working with
- We will use VMD for this
 - Made by the same developers as NAMD
 - Rasmus will intro us on the details
- VMD is installed on Mahti and Puhti; You can also use the web interface of both for visualization
 - puhti.csc.fi & mahti.csc.fi
 - Although this is not advisable, as the graphics output can be quite slow there

